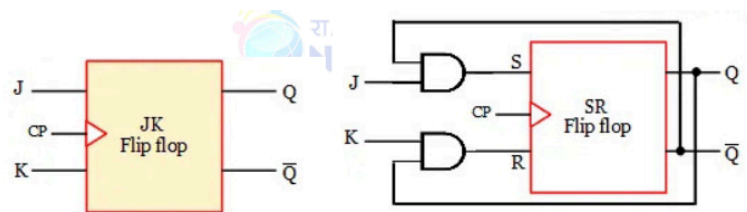


# Circuit Simulation Lab:13

## J-K Flip-flops

The purpose of this experiment is to introduce the design of simple sequential circuits, in this case J-K Flipflop. JK flip flop is a sequential bi-state single-bit memory element. It has two inputs (J and K), two outputs(Q and Q') and a clock pulse input. It can be triggered either at the positive edge or at the negative edge of the clock pulse.



**J-K Flip Flop**

| CP | J | K | $Q_{n+1}$   | State     |
|----|---|---|-------------|-----------|
| 1  | 0 | 0 | $Q_n$       | NO CHANGE |
| 1  | 0 | 1 | 0           | RESET     |
| 1  | 1 | 0 | 1           | SET       |
| 1  | 1 | 1 | $\bar{Q}_n$ | TOGGLES   |

### Behavior modeling

JK FF:

```
module jk(  
input clk , j , k ,  
output reg q);  
  
always @(posedge clk)begin  
  
q <= ((j & (~q)) | (~k & q ));  
end  
endmodule
```

Testbench

```
module tb_jk_ff;
```

```
reg j,k,clk;
```

```
wire q;
```

```
jk    dut(
```

```
  .j(j),
```

```
  .k(k),
```

```
  .clk(clk),
```

```
  .q(q)
```

```
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
  $dumpfile("jk_ff.vcd");
```

```
  $dumpvars(0,tb_jk_ff);
```

```
  clk = 0;
```

```
  j = 0; k = 0; #20;
```

```
  j = 0; k = 1; #20;
```

```
  j = 1; k = 0; #20;
```

```
  j = 1; k = 1; #20;
```

```
  j = 0; k = 0; #20;
```

```
  j = 1; k = 1; #20;
```

```
  j = 1; k = 0; #20;
```

```
j = 0; k = 1; #20;
```

```
$finish;
```

```
end
```

```
initial begin
```

```
$monitor("time=%0t clk=%b j=%b k=%b q=%b",
```

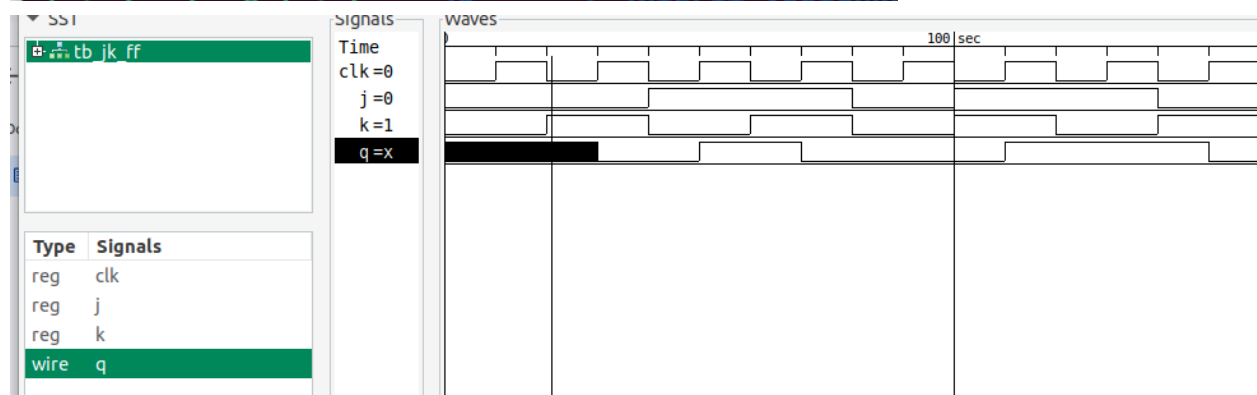
```
    $time,clk,j,k,q);
```

```
end
```

```
Endmodule
```

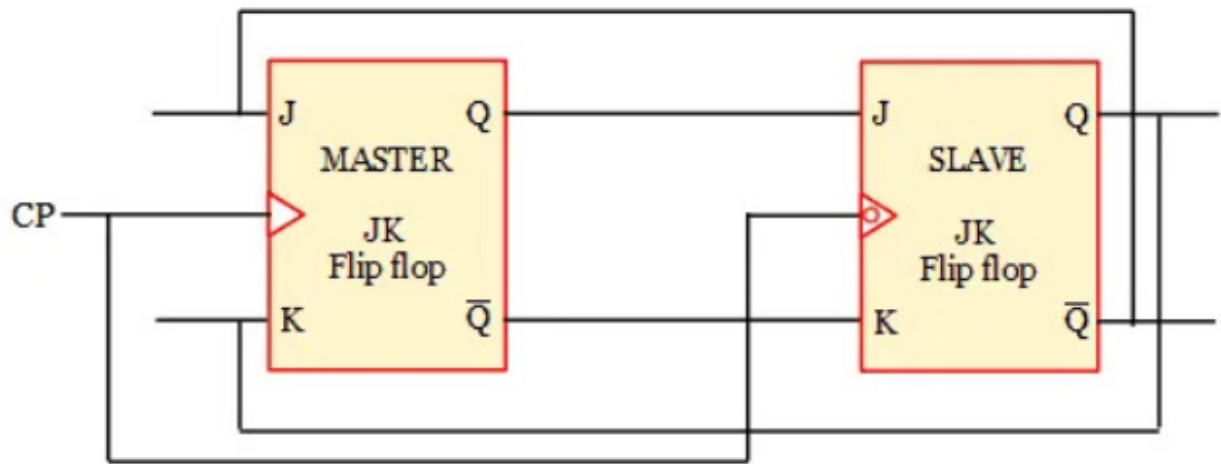
## Outputs

```
VCD info: dumpfile jk_ff.vcd opened for output.  
time=0  clk=0  j=0  k=0  q=x  
time=10  clk=1  j=0  k=0  q=x  
time=20  clk=0  j=0  k=1  q=x  
time=30  clk=1  j=0  k=1  q=0  
time=40  clk=0  j=1  k=0  q=0  
time=50  clk=1  j=1  k=0  q=1  
time=60  clk=0  j=1  k=1  q=1  
time=70  clk=1  j=1  k=1  q=0  
time=80  clk=0  j=0  k=0  q=0  
time=90  clk=1  j=0  k=0  q=0  
time=100 clk=0  j=1  k=1  q=0  
time=110 clk=1  j=1  k=1  q=1  
time=120 clk=0  j=1  k=0  q=1  
time=130 clk=1  j=1  k=0  q=1  
time=140 clk=0  j=0  k=1  q=1  
time=150 clk=1  j=0  k=1  q=0  
time=160 clk=0  j=0  k=1  q=0
```



## Structural modeling (Use the JK Flip Flop for designing the MS-JK)

### Master Slave JK Flip-Flop



```
CODE:
module ms_jk(
input clk,
input j,
input k,
output q
);

wire qm;
wire clk_bar;

not n1(clk_bar,clk);

jk master(
.clk(clk),
.j(j),
.k(k),
.q(qm)
);

jk slave(
.clk(clk_bar),
```

```

.j(qm),
.k(~qm),
.q(q)
);

Endmodule

Testbench
module tb_ms_jk;

reg clk,j,k;
wire q;

ms_jk dut(
.clk(clk),
.j(j),
.k(k),
.q(q)
);

always #10 clk = ~clk;

initial begin

$dumpfile("ms_jk.vcd");
$dumpvars(0,tb_ms_jk);

clk = 0;

j=0; k=0; #20;

j=0; k=1; #20;

j=1; k=0; #20;

j=1; k=1; #20;

j=0; k=0; #20;

```

```
j=1; k=1; #20;
```

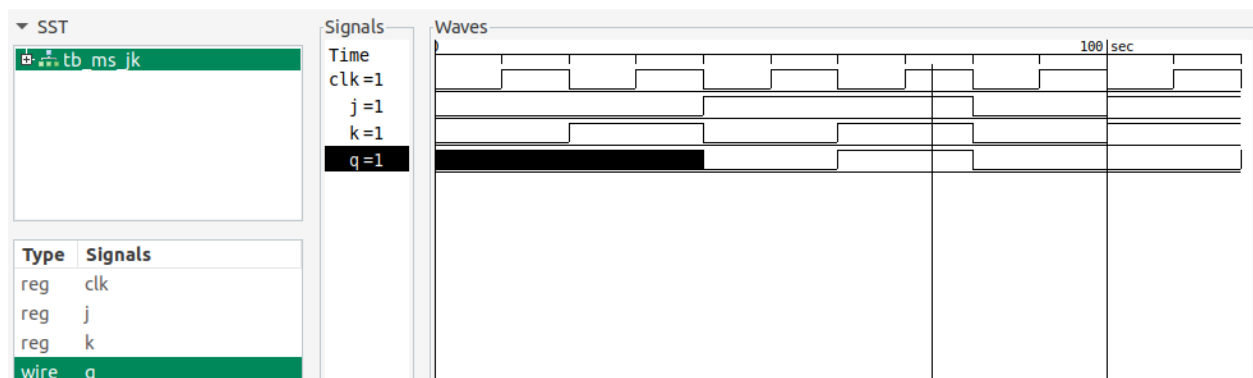
```
$finish;
```

```
end
```

```
initial
```

```
$monitor("t=%0t clk=%b j=%b k=%b q=%b",  
        $time,clk,j,k,q);
```

```
endmodule
```



```
VCD info: dumpfile ms_jk.vcd opened for output.  
t=0  clk=0  j=0  k=0  q=x  
t=10 clk=1  j=0  k=0  q=x  
t=20 clk=0  j=0  k=1  q=x  
t=30 clk=1  j=0  k=1  q=x  
t=40 clk=0  j=1  k=0  q=0  
t=50 clk=1  j=1  k=0  q=0  
t=60 clk=0  j=1  k=1  q=1  
t=70 clk=1  j=1  k=1  q=1  
t=80 clk=0  j=0  k=0  q=0  
t=90 clk=1  j=0  k=0  q=0  
t=100 clk=0 j=1  k=1  q=0  
t=110 clk=1 j=1  k=1  q=0  
t=120 clk=0 j=1  k=1  q=1
```